

Zen Nguyen

(672) 673-2613 | ttn54@sfu.ca | linkedin.com/in/zennguyen1305 | github.com/ttn54 | zennguyen.me

TECHNICAL SKILLS

- **Backend & Systems:** FastAPI, gRPC, RESTful APIs
- **Infrastructure & DevOps:** Docker, Linux, AWS (EC2)
- **AI & Data:** RAG Architecture, Vector Databases, LangChain, OpenAI API, Gemini API
- **Languages:** Python, Go, Java, C, C++, TypeScript, JavaScript, SQL, HTML/CSS

TECHNICAL PROJECTS

TitanSwarm (Autonomous AI Co-Pilot)

Jan 2026 - Present

Personal Project · Python, LangChain, RAG Architecture, Gemini API

- Automated end-to-end job application workflows by architecting an autonomous AI Co-Pilot using **Python** and **LangChain**, resulting in the discovery, extraction, and parsing of real-time Software Engineering postings.
- Delivered uniquely tailored, ATS-optimized PDF resumes by building a zero-hallucination **RAG Architecture** engine using **Gemini API** and a FAISS Vector Store, processing verified user data.
- Enabled users to review and manage applications across a full pipeline by designing and implementing a Streamlit Human-in-the-Loop dispatch UI with a Kanban board and persistent state via a SQLite repository.
- Validated all components including the scraper, AI tailor, PDF generator, and repository layer by applying Test-Driven Development (TDD) with a 113-test suite using Pytest, mocking, and async fixtures.

TitanStore (Distributed Raft Database)

Jan 2026 - Present

Personal Project · Go, Raft Consensus Algorithm, TCP Sockets, WAL

- Ensured data durability and enabled node recovery after failures without data loss by engineering a crash-safe Write-Ahead Log (WAL) with atomic snapshots.
- Developed a distributed key-value database in **Go** from scratch, implementing the **Raft consensus** protocol for leader election and fault-tolerant log replication across a cluster of nodes.
- Designed and implemented a high-throughput raw **TCP Sockets** server on port 6001, parsing a custom text protocol supporting SET, GET, and ERR NOT_LEADER operations with non-blocking I/O.

SFU Course Tracker (sfucourseplanner.me)

Nov 2025 - Jan 2026

Personal Project · Python, Docker, AWS (EC2), FastAPI

- Optimized deployments and reduced infrastructure costs by migrating the full-stack platform from Azure to **AWS (EC2)**, implementing **Docker** container orchestration.
- Increased data throughput by 10x while respecting server constraints by engineering a scraping pipeline using Asyncio and HTTPX with semaphore-based rate limiting.
- Designed and implemented a custom parser to tokenize and evaluate nested boolean prerequisite strings, converting unstructured text into a deterministic Abstract Syntax Tree (AST).

EDUCATION

Bachelor of Science, Computing Science

May 2025 - Present

Simon Fraser University

- CGPA: 3.74 / 4.33
- Relevant Coursework: Discrete Math (A+), Computer Systems(A), Data Structures & Programming (B+), Linear Algebra(A-).
- Langara College | Associate of Science, Computer Science Jan 2024 - Apr 2025